27高等数学17堂课

专题13　多元函数微分法

(P130-138)

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（一）复合函数求导法

设 $u=u(x,y), v=v(x,y)$ 可导，$z=f(u,v)$ 在相应点有连续一阶偏导数，则

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial y}$$

（二）全微分形式不变性

设 $z=f(u,v), u=u(x,y), v=v(x,y)$ 都有连续一阶偏导数，则

$$dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy$$

$$dz=\frac{\partial z}{\partial u}du+\frac{\partial z}{\partial v}dv$$



(三) 隐函数求导法

1. 由一个方程所确定的隐函数

设 $F(x,y,z)$ 有连续一阶偏导数，$F_z \neq 0$，$z=z(x,y)$ 由 $F(x,y,z)=0$ 所确定.

方法：

(1) 公式 $\dfrac{\partial z}{\partial x}=-\dfrac{F_x}{F_z}$，$\dfrac{\partial z}{\partial y}=-\dfrac{F_y}{F_z}$；

(2) 等式两边求导

$$F_x+F_z\frac{\partial z}{\partial x}=0,\quad F_y+F_z\frac{\partial z}{\partial y}=0,$$

(3) 利用微分形式不变性

$$F_x dx+F_y dy+F_z dz=0$$

2. 由方程组所确定的隐函数（仅数一要求）

设 $u=u(x,y), v=v(x,y)$ 由 $\begin{cases} F(x,y,u,v)=0 \\ G(x,y,u,v)=0 \end{cases}$ 所确定.

方法：

（1）等式两边求导

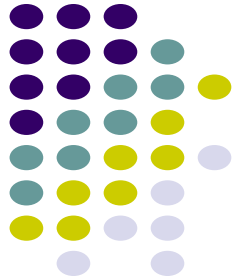
$$\begin{cases} F_x + F_u \dfrac{\partial u}{\partial x} + F_v \dfrac{\partial v}{\partial x} = 0 \\ G_x + G_u \dfrac{\partial u}{\partial x} + G_v \dfrac{\partial v}{\partial x} = 0 \end{cases}$$

（2）微分形式不变性

$$\begin{cases} F_x dx + F_y dy + F_u du + F_v dv = 0 \\ G_x dx + G_y dy + G_u du + G_v dv = 0 \end{cases}$$

1. 复合函数偏导数与全微分

【例1】设 $z=\dfrac{x\cos(y-1)-(y-1)\cos x}{1+\sin x+\sin(y-1)}$，则 $\left.\dfrac{\partial z}{\partial y}\right|_{(0,1)}=$ _____ .



【例2】设函数 $z=(1+\frac{x}{y})^{\frac{x}{y}}$，则 $dz|_{(1,1)}=$ _____ .

$(1+2\ln 2)(dx-dy)$

【解1】 $z(x,1)=[(1+x)^x]=e^{x\ln(1+x)}$

$z_x(x,1)=e^{x\ln(1+x)}\left[\ln(1+x)+\frac{x}{1+x}\right]$ $z_x(1,1)=2\left[\ln 2+\frac{1}{2}\right]=1+2\ln 2$

$z(1,y)=(1+y)^{\frac{1}{y}}$ $\frac{1}{y}=u$ $(1+u)^u$, $z_y(1,1)=\left[(1+2)^{\frac{1}{2}}\ln 2\right](-1)=-(1+2\ln 2)$

【解2】 $z=(1+\frac{x}{y})^{\frac{x}{y}}=(1+u)^{\frac{1}{u}}=e^{\frac{1}{u}\ln(1+u)}$ $\left(u=\frac{x}{y}\right)$

$dz=z_u\,du=e^{\frac{1}{u}\ln(1+u)}\left[-\frac{1}{u^2}\ln(1+u)+\frac{1}{u(1+u)}\right]\left(\frac{x}{y}\right)\frac{y\,dx-x\,dy}{y^2}$

$dz|_{(1,1)}=2\left[-\ln 2+\frac{1}{2}\right](dx-dy)$

【例3】设函数 $F(x,y)=\int_0^{xy}\frac{\sin t}{1+t^2}\mathrm{d}t$，则

$$\left.\frac{\partial^2 F}{\partial x^2}\right|_{\substack{x=0\\y=2}}=_____.$$

【解1】 $$\frac{\partial F}{\partial x}=\frac{y\sin xy}{1+x^2y^2}$$

$$\frac{\partial^2 F}{\partial x^2}=\frac{y^2\cos(xy)(1+x^2y^2)-2xy^3\sin xy}{(1+x^2y^2)^2}$$

故 $$\left.\frac{\partial^2 F}{\partial x^2}\right|_{\substack{x=0\\y=2}}=4.$$

【例3】设函数 $F(x,y)=\int_0^{xy}\frac{\sin t}{1+t^2}\mathrm{d}t$，则

$\left.\frac{\partial^2 F}{\partial x^2}\right|_{\substack{x=0\\y=2}}=$ _____ .

$\varphi'(0)$

$\varphi(0)=0$

【解2】 $\frac{\partial F}{\partial x}=\frac{y\sin xy}{1+x^2y^2}$

$F_x(x,2)=\frac{2\sin 2x}{1+4x^2}=\varphi(x)$

$\left.\frac{\partial^2 F}{\partial x^2}\right|_{\substack{x=0\\y=2}}=F_{xx}(0,2)=\lim_{x\to 0}\frac{2\sin 2x}{x(1+4x^2)}$

$=\lim_{x\to 0}\frac{4x}{x(1+4x^2)}=4$

① 复合求导

② 对y

【例4】设 $z=(1+x^2y^2)^{xy}$，求 $\dfrac{\partial z}{\partial x}$ 及 $\dfrac{\partial z}{\partial y}$.

【解1】 $z=e^{xy\ln(1+x^2y^2)}$

【解2】 $\ln z=xy\ln(1+x^2y^2)$

【解3】 令 $u=1+x^2y^2, v=xy$, 则 $z=u^v$

$$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial x}=vu^{v-1}2xy^2+u^v\ln u\cdot y$$

$$=(1+x^2y^2)^{xy}\left[\frac{2x^2y^3}{1+x^2y^2}+y\ln(1+x^2y^2)\right]$$

同理可得 $\dfrac{\partial z}{\partial y}$.

【例5】设函数 $f(u,v)$ 满足 $f(x+y,\frac{y}{x})=x^2-y^2$，则 $\left.\frac{\partial f}{\partial u}\right|_{\substack{u=1\\v=1}}$ 与 $\left.\frac{\partial f}{\partial v}\right|_{\substack{u=1\\v=1}}$ 依次是（ ）

（A）$\frac{1}{2},0.$　　（B）$0,\frac{1}{2}.$　　（C）$-\frac{1}{2},0.$　　（D）$0,-\frac{1}{2}.$

【解1】令 $x+y=u,\frac{y}{x}=v,$

则 $x=\frac{u}{1+v},y=\frac{uv}{1+v}$

故 $f(u,v)=\left(\frac{u}{1+v}\right)^2-\left(\frac{uv}{1+v}\right)^2=\frac{u^2(1-v)}{1+v}$

所以 $\frac{\partial f}{\partial u}=\frac{2u(1-v)}{1+v},\quad \frac{\partial f}{\partial v}=-\frac{2u^2}{(1+v)^2}$

$\left.\frac{\partial f}{\partial u}\right|_{\substack{u=1\\v=1}}=0,\quad \left.\frac{\partial f}{\partial v}\right|_{\substack{u=1\\v=1}}=-\frac{1}{2}$

【例5】设函数 $f(u,v)$ 满足 $f(x+y,\frac{y}{x})=x^2-y^2$，则 $\left.\frac{\partial f}{\partial u}\right|_{\substack{u=1\\v=1}}$ 与 $\left.\frac{\partial f}{\partial v}\right|_{\substack{u=1\\v=1}}$ 依次是（ ）

（A）$\frac{1}{2},0.$　　（B）$0,\frac{1}{2}.$　　（C）$-\frac{1}{2},0.$　　（D）$0,-\frac{1}{2}.$

【解2】令 $x+y=u,\frac{y}{x}=v,$ 则当 $u=1,v=1$ 时，$x=y=\frac{1}{2}.$

$$f_u+f_v(-\frac{y}{x^2})=2x$$

$$f_u+f_v(\frac{1}{x})=-2y$$

将 $x=y=\frac{1}{2}$ 代入上式得 $\begin{cases} f_u(1,1)-2f_v(1,1)=1 \\ f_u(1,1)+2f_v(1,1)=-1 \end{cases}$

由此解得 $f_u(1,1)=0, f_v(1,1)=-\frac{1}{2}.$

【例6】设函数 $z=f(x,y)$ 在点 $(1,1)$ 处可微, 且 $f(1,1)=1$,

$$\left.\frac{\partial f}{\partial x}\right|_{(1,1)}=2,\left.\frac{\partial f}{\partial y}\right|_{(1,1)}=3,\quad \varphi(x)=f(x,f(x,x)).\quad 求\left.\frac{\mathrm{d}}{\mathrm{d}x}\varphi^3(x)\right|_{x=1}.$$

【解】 $\varphi(1)=f(1,f(1,1))=f(1,1)=1,$

$$\left.\frac{\mathrm{d}}{\mathrm{d}x}\varphi^3(x)\right|_{x=1}=\left[3\varphi^2(x)\frac{\mathrm{d}\varphi(x)}{\mathrm{d}x}\right]_{x=1}$$

$$=3\varphi^2(x)[f_1'(x,f(x,x))+f_2'(x,f(x,x))(f_1'(x,x)+f_2'(x,x))]\Big|_{x=1}$$

$$=3\cdot1\cdot[2+3(2+3)]=51.$$

【例7】设 $z=x^3 f(xy,\frac{y}{x})$，f 具有连续二阶偏导数，求 $\frac{\partial z}{\partial y},\frac{\partial^2 z}{\partial y^2}$ 及 $\frac{\partial^2 z}{\partial x\partial y}$.

【解】 $\frac{\partial z}{\partial y}=x^4 f_1'+x^2 f_2',$

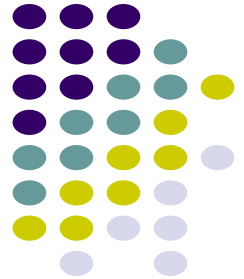
$$\frac{\partial^2 z}{\partial y^2}=x^4\left[xf_{11}''+\frac{1}{x}f_{12}''\right]+x^2\left[xf_{21}''+\frac{1}{x}f_{22}''\right]$$

$$=x^5 f_{11}''+2x^3 f_{12}''+xf_{22}'',$$

$$\frac{\partial^2 z}{\partial x\partial y}=4x^3 f_1'+x^4\left[yf_{11}''-\frac{y}{x^2}f_{12}''\right]+2xf_2'+x^2\left[yf_{21}''-\frac{y}{x^2}f_{22}''\right]$$

$$=4x^3 f_1'+2xf_2'+x^4 yf_{11}''-yf_{22}''.$$

【例8】设 $u=f(x,y,z), z=\int_0^{x+y} e^{-t^2}dt$. 求 $\dfrac{\partial u}{\partial x}, \dfrac{\partial^2 u}{\partial x\partial y}$，其中 f 有二阶连续偏导数.



[解] $\dfrac{\partial u}{\partial x}=\dfrac{\partial f}{\partial x}+\dfrac{\partial f}{\partial z}e^{-(x+y)^2}$

$\dfrac{\partial^2 u}{\partial x\partial y}=\dfrac{\partial^2 f}{\partial x\partial y}+\dfrac{\partial^2 f}{\partial x\partial z}e^{-(x+y)^2}+e^{-(x+y)^2}\left[-2(x+y)\right]\dfrac{\partial f}{\partial z}$

$+e^{-(x+y)^2}\left[\dfrac{\partial^2 f}{\partial z\partial y}+\dfrac{\partial^2 f}{\partial z^2}e^{-(x+y)^2}\right]$

$u=f(x,y,z)$

$x\ y\ z$

【例9】设函数 $z=f(xy,yg(x))$，其中函数 f 具有二阶连续偏导数，函数 $g(x)$ 可导且在 $x=1$ 处取得极值 $g(1)=1$. 求 $\left.\dfrac{\partial^2 z}{\partial x\partial y}\right|_{\substack{x=1\\y=1}}$.

【解1】由 $z=f(xy,yg(x))$ 知

$$\frac{\partial z}{\partial x}=yf_1'+yg'(x)f_2',$$

$$\frac{\partial^2 z}{\partial x\partial y}=f_1'+y[xf_{11}''+g(x)f_{12}'']+g'(x)f_2'+yg'(x)[xf_{21}''+g(x)f_{22}''].$$

由题意 $g(1)=1, g'(1)=0$，在上式中令 $x=1, y=1$ 得

$$\left.\frac{\partial^2 z}{\partial x\partial y}\right|_{\substack{x=1\\y=1}}=f_1'(1,1)+f_{11}''(1,1)+f_{12}''(1,1).$$

【例9】设函数 $z=f(xy,yg(x))$，其中函数 f 具有二阶连续偏导数，函数 $g(x)$ 可导且在 $x=1$ 处取得极值 $g(1)=1$. 求 $\left.\dfrac{\partial^2 z}{\partial x\partial y}\right|_{\substack{x=1\\y=1}}$.

【解2】由 $z=f(xy,yg(x))$ 知

$$\frac{\partial z}{\partial x}=yf_1'+yg'(x)f_2',$$

由题意 $g(1)=1,g'(1)=0,$ 在上式中令 $x=1$ 得

$$z_x(1,y)=yf_1'(y,y)$$

$$z_{xy}(1,y)=f_1'(y,y)+y[f_{11}''(y,y)+f_{12}''(y,y)]$$

$$\left.\frac{\partial^2 z}{\partial x\partial y}\right|_{\substack{x=1\\y=1}}=f_1'(1,1)+f_{11}''(1,1)+f_{12}''(1,1).$$

【例10】设变换 $\begin{cases} u = x - 2y, \\ v = x + ay \end{cases}$ 可把方程 $6\dfrac{\partial^2 z}{\partial x^2} + \dfrac{\partial^2 z}{\partial x \partial y} - \dfrac{\partial^2 z}{\partial y^2} = 0$

简化为 $\dfrac{\partial^2 z}{\partial u \partial v} = 0$, 求常数 a.

【解1】将 z 视为以 u, v 为中间变量的 x, y 的复合函数, 由题设可得

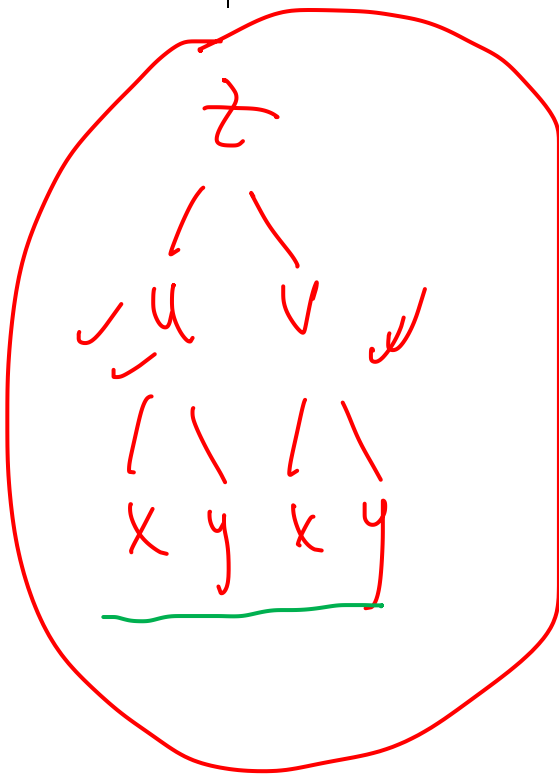
$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}, \quad \frac{\partial z}{\partial y} = -2\frac{\partial z}{\partial u} + a\frac{\partial z}{\partial v},$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial u^2} + 2\frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}, \quad \frac{\partial^2 z}{\partial y^2} = 4\frac{\partial^2 z}{\partial u^2} - 4a\frac{\partial^2 z}{\partial u \partial v} + a^2\frac{\partial^2 z}{\partial v^2},$$

$$\frac{\partial^2 z}{\partial x \partial y} = -2\frac{\partial^2 z}{\partial u^2} + (a-2)\frac{\partial^2 z}{\partial u \partial v} + a\frac{\partial^2 z}{\partial v^2}.$$

$$(10 + 5a)\frac{\partial^2 z}{\partial u \partial v} + (6 + a - a^2)\frac{\partial^2 z}{\partial v^2} = 0.$$

$6 + a - a^2 = 0$ 且 $10 + 5a \neq 0$, 解之得 $a = 3$.



【例10】设变换 $\begin{cases} u = x - 2y, \\ v = x + ay \end{cases}$ 可把方程 $6\dfrac{\partial^2 z}{\partial x^2} + \dfrac{\partial^2 z}{\partial x \partial y} - \dfrac{\partial^2 z}{\partial y^2} = 0$

简化为 $\dfrac{\partial^2 z}{\partial u \partial v} = 0$, 求常数 a.

【解2】将 z 视为以 x, y 为中间变量的 u, v 的复合函数, 由题设可得



$$x = \frac{au + 2v}{a+2}, \quad y = \frac{-u + v}{a+2}.$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u} = \frac{a}{a+2}\cdot\frac{\partial z}{\partial x} - \frac{1}{a+2}\frac{\partial z}{\partial y},$$

$$\frac{\partial^2 z}{\partial u \partial v} = \frac{a}{a+2}\left(\frac{\partial^2 z}{\partial x^2}\frac{\partial x}{\partial v} + \frac{\partial^2 z}{\partial x \partial y}\frac{\partial y}{\partial v}\right) - \frac{1}{a+2}\left(\frac{\partial^2 z}{\partial y \partial x}\frac{\partial x}{\partial v} + \frac{\partial^2 z}{\partial y^2}\frac{\partial y}{\partial v}\right)$$

$$= \frac{2a}{(a+2)^2}\frac{\partial^2 z}{\partial x^2} + \frac{a-2}{(a+2)^2}\frac{\partial^2 z}{\partial x \partial y} - \frac{1}{(a+2)^2}\frac{\partial^2 z}{\partial y^2}. \quad = 0$$

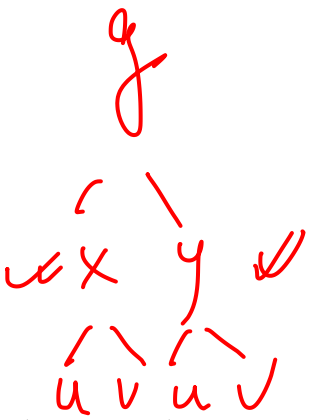
$$2a\frac{\partial^2 z}{\partial x^2} + (a-2)\frac{\partial^2 z}{\partial x \partial y} - \frac{\partial^2 z}{\partial y^2} = 0. \qquad \frac{2a}{6} = \frac{a-2}{1} = \frac{-1}{-1} \Rightarrow a = 3.$$

【例11】（2024年2）已知 $f(u,v)$ 具有二阶连续偏导数，且 $g(x,y)=f(2x+y,3x-y)$

满足 $\dfrac{\partial^2 g}{\partial x^2}+\dfrac{\partial^2 g}{\partial x\partial y}-6\dfrac{\partial^2 g}{\partial y^2}=1.$

（I）求 $\dfrac{\partial^2 f}{\partial u\partial v}$；

（II）若 $\dfrac{\partial f(u,0)}{\partial u}=ue^{-u}$，$f(0,v)=\dfrac{1}{50}v^2-1$，求 $f(u,v)$ 的表达式.



【解】（I）令 $u=2x+y, v=3x-y$，则 $f(u,v)=g(x,y)$，其中 $x=\dfrac{1}{5}u+\dfrac{1}{5}v, y=\dfrac{3}{5}u-\dfrac{2}{5}v.$

$$\frac{\partial f}{\partial u}=\frac{\partial g}{\partial x}\frac{\partial x}{\partial u}+\frac{\partial g}{\partial y}\frac{\partial y}{\partial u}=\frac{1}{5}\frac{\partial g}{\partial x}+\frac{3}{5}\frac{\partial g}{\partial y}$$

$$\frac{\partial^2 f}{\partial u\partial v}=\frac{1}{5}\left[\frac{\partial^2 g}{\partial x^2}\frac{\partial x}{\partial v}+\frac{\partial^2 g}{\partial x\partial y}\frac{\partial y}{\partial v}\right]+\frac{3}{5}\left[\frac{\partial^2 g}{\partial y\partial x}\frac{\partial x}{\partial v}+\frac{\partial^2 g}{\partial y^2}\frac{\partial y}{\partial v}\right]$$

$$=\frac{1}{5}\left[\frac{\partial^2 g}{\partial x^2}\frac{1}{5}-\frac{\partial^2 g}{\partial x\partial y}\frac{2}{5}\right]+\frac{3}{5}\left[\frac{\partial^2 g}{\partial y\partial x}\frac{1}{5}-\frac{\partial^2 g}{\partial y^2}\frac{2}{5}\right]=\frac{1}{25}\left[\frac{\partial^2 g}{\partial x^2}+\frac{\partial^2 g}{\partial x\partial y}-6\frac{\partial^2 g}{\partial y^2}\right]=\frac{1}{25}$$

（II）由 $\dfrac{\partial^2 f}{\partial u \partial v}=\dfrac{1}{25}$ 可知 $\dfrac{\partial f}{\partial u}=\int \dfrac{1}{25}dv=\dfrac{1}{25}v+\varphi(u)$

由 $\dfrac{\partial f(u,0)}{\partial u}=ue^{-u}$ 可知，$\varphi(u)=ue^{-u}$，即 $\dfrac{\partial f}{\partial u}=\dfrac{1}{25}v+ue^{-u}$，则

$$f(u,v)=\int[\frac{1}{25}v+ue^{-u}]du=\frac{1}{25}uv-ue^{-u}-e^{-u}+\psi(v)$$

又 $f(0,v)=\dfrac{1}{50}v^2-1$，则 $\psi(v)=\dfrac{1}{50}v^2$

$$f(u,v)=\frac{1}{25}uv+\frac{1}{50}v^2-(u+1)e^{-u}.$$

【例12】已知函数 $z=z(x,y)$ 满足 $x^2\frac{\partial z}{\partial x}+y^2\frac{\partial z}{\partial y}=z^2$. 设 $u=x, v=\frac{1}{y}-\frac{1}{x}, \psi=\frac{1}{z}-\frac{1}{x}$ 对函数 $\psi=\psi(u,v)$, 求证 $\frac{\partial \psi}{\partial u}=0.$

【证】 由 $\begin{cases} u=x \\ v=\frac{1}{y}-\frac{1}{x}, \end{cases}$ 解得 $\begin{cases} x=u, \\ y=\frac{u}{1+uv}. \end{cases}$

$$\frac{\partial \psi}{\partial u}=-\frac{1}{z^2}\left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial u}\right)+\frac{1}{u^2}$$

$$=-\frac{1}{z^2}\left(\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}\frac{1}{(1+uv)^2}\right)+\frac{1}{u^2},$$

$$\frac{\partial \psi}{\partial u}=-\frac{1}{z^2x^2}\left(x^2\frac{\partial z}{\partial x}+y^2\frac{\partial z}{\partial y}\right)+\frac{1}{u^2}=-\frac{1}{x^2}+\frac{1}{u^2}=0.$$

2. 隐函数的偏导数与全微分

【例1】若函数 $z=z(x,y)$ 由方程 $e^{x+2y+3z}+xyz=1$ 确定，则

$dz\big|_{(0,0)}=$ _____ .

【解1】由 $x=0,y=0$ 知 $z=0$

方程 $e^{x+2y+3z}+xyz=1$ 两端微分得

$$e^{x+2y+3z}(dx+2dy+3dz)+yzdx+xzdy+xydz=0$$

将 $x=0,y=0,z=0$ 代入上式得

$$dx+2dy+3dz=0$$

则 $$dz\big|_{(0,0)}=-\frac{1}{3}(dx+2dy).$$

【例1】若函数 $z=z(x,y)$ 由方程 $e^{x+2y+3z}+xyz=1$ 确定，则

$dz|_{(0,0)}=$ _____.

【解2】由 $x=0, y=0$ 知 $z=0$

$$dz|_{(0,0)}=z_x(0,0)dx+z_y(0,0)dy$$

在 $e^{x+2y+3z}+xyz=1$ 中令 $y=0$ 得，$e^{x+3z}=1$, 两边对 x 求导得

$$e^{x+3z}(1+3z_x)=0$$

$$z_x(0,0)=-\frac{1}{3}$$

同理可得 $z_y(0,0)=-\frac{2}{3}$

则 $dz|_{(0,0)}=-\frac{1}{3}(dx+2dy).$

【例2】设函数 $z=z(x,y)$ 由方程 $F\left(\frac{y}{x},\frac{z}{x}\right)=0$ 确定，其中 F 为可微函数，且 $F_2'\neq 0$，则 $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=$（　）.

（A）x　　（B）z　　（C）$-x$　　（D）$-z$

【解】 $\frac{\partial z}{\partial x}=-\frac{-\frac{y}{x^2}F_1-\frac{z}{x^2}F_2}{\frac{1}{x}F_2}$，$\frac{\partial z}{\partial y}=-\frac{\frac{1}{x}F_1}{\frac{1}{x}F_2}$，

$$x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=\frac{-\frac{y}{x}F_1-\frac{z}{x}F_2}{\frac{1}{x}F_2}-\frac{\frac{y}{x}F_1}{\frac{1}{x}F_2}=z$$

故应选（B）.

【例3】设 $u=f(x,y,z)$ 有连续的一阶偏导数，又函数 $y=y(x)$ 及 $z=z(x)$ 分别由下列两式确定：$\mathrm{e}^{xy}-xy=2$ 和 $\mathrm{e}^{x}=\int_0^{x-z}\frac{\sin t}{t}\mathrm{d}t$，求 $\frac{\mathrm{d}u}{\mathrm{d}x}$.

【解1】
$$\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{\partial f}{\partial z}\frac{\mathrm{d}z}{\mathrm{d}x}. \qquad (1)$$

由 $\mathrm{e}^{xy}-xy=2$ 两边对 x 求导，得

$$\mathrm{e}^{xy}\left(y+x\frac{\mathrm{d}y}{\mathrm{d}x}\right)-\left(y+x\frac{\mathrm{d}y}{\mathrm{d}x}\right)=0,\qquad \frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{y}{x}.$$

又由 $\mathrm{e}^{x}=\int_0^{x-z}\frac{\sin t}{t}\mathrm{d}t$ 两边对 x 求导，得

$$\mathrm{e}^{x}=\frac{\sin(x-z)}{x-z}\cdot\left(1-\frac{\mathrm{d}z}{\mathrm{d}x}\right),\qquad \frac{\mathrm{d}z}{\mathrm{d}x}=1-\frac{\mathrm{e}^{x}(x-z)}{\sin(x-z)}.$$

$$\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{\partial f}{\partial x}-\frac{y}{x}\frac{\partial f}{\partial y}+\left[1-\frac{\mathrm{e}^{x}(x-z)}{\sin(x-z)}\right]\frac{\partial f}{\partial z}.$$

【例3】设 $u=f(x,y,z)$ 有连续的一阶偏导数，又函数 $y=y(x)$ 及 $z=z(x)$ 分别由下列两式确定：$e^{xy}-xy=2$ 和 $e^{x}=\int_0^{x-z}\frac{\sin t}{t}dt$，求 $\frac{du}{dx}$.

【解2】 $du=\frac{\partial f}{\partial x}dx+\frac{\partial f}{\partial y}dy+\frac{\partial f}{\partial z}dz$ (1)

等式 $e^{xy}-xy=2$ 两端微分得

$$e^{xy}(ydx+xdy)-(ydx+xdy)=0,\quad dy=-\frac{y}{x}dx$$

等式 $e^{x}=\int_0^{x-z}\frac{\sin t}{t}dt$ 两端微分得

$$e^{x}dx=\frac{\sin(x-z)}{x-z}(dx-dz)\qquad dz=(1-\frac{e^{x}(x-z)}{\sin(x-z)})dx.$$

$$du=[\frac{\partial f}{\partial x}-\frac{y}{x}\frac{\partial f}{\partial y}+\left[1-\frac{e^{x}(x-z)}{\sin(x-z)}\right]\frac{\partial f}{\partial z}]dx$$

【例4】设 $z=z(x,y)$ 是由方程 $x^2+y^2-z=\varphi(x+y+z)$ 所确定的函数，其中 φ 具有二阶导数，且 $\varphi'\neq -1$.

（I）求 $\mathrm{d}z$

（II）记 $u(x,y)=\dfrac{1}{x-y}\left(\dfrac{\partial z}{\partial x}-\dfrac{\partial z}{\partial y}\right)$，求 $\dfrac{\partial u}{\partial x}$

【解1】（I）设 $F(x,y,z)=x^2+y^2-z-\varphi(x+y+z)$，则

$$\frac{\partial z}{\partial x}=-\frac{F_x'}{F_z'}=\frac{2x-\varphi'}{1+\varphi'}\qquad \frac{\partial z}{\partial y}=-\frac{F_y'}{F_z'}=\frac{2y-\varphi'}{1+\varphi'}$$

$$\mathrm{d}z=\frac{\partial z}{\partial x}\mathrm{d}x+\frac{\partial z}{\partial y}\mathrm{d}y=\frac{1}{1+\varphi'}[(2x-\varphi')\mathrm{d}x+(2y-\varphi')\mathrm{d}y].$$

（II）由于 $u(x,y)=\dfrac{2}{1+\varphi'}$，所以

$$\frac{\partial u}{\partial x}=\frac{-2}{(1+\varphi')^2}\left(1+\frac{\partial z}{\partial x}\right)\varphi''=-\frac{2(2x+1)\varphi''}{(1+\varphi')^3}.$$

【例4】设 $z=z(x,y)$ 是由方程 $x^2+y^2-z=\varphi(x+y+z)$ 所确定的函数, 其中 φ 具有二阶导数，且 $\varphi'\neq -1$.

（I）求 $\mathrm{d}z$

（II）记 $u(x,y)=\dfrac{1}{x-y}\left(\dfrac{\partial z}{\partial x}-\dfrac{\partial z}{\partial y}\right)$，求 $\dfrac{\partial u}{\partial x}$

【解2】（I）对等式 $x^2+y^2-z=\varphi(x+y+z)$ 两端求微分，得

$$2x\,\mathrm{d}x+2y\,\mathrm{d}y-\mathrm{d}z=\varphi'\cdot(\mathrm{d}x+\mathrm{d}y+\mathrm{d}z).$$

解出 $\mathrm{d}z$，得

$$\mathrm{d}z=\frac{2x-\varphi'}{1+\varphi'}\mathrm{d}x+\frac{2y-\varphi'}{1+\varphi'}\mathrm{d}y.$$

(II) 同解1.



【例5】设 $z=z(x,y)$ 是由方程 $f(y-x,yz)=0$ 所确定的隐函数，

其中函数 f 对各个变量具有连续的二阶偏导数，求 $\dfrac{\partial z}{\partial x}$ 及 $\dfrac{\partial^2 z}{\partial x^2}$.

【解】方程 $f(y-x,yz)=0$ 的两边对 x 求导，得

$$-f_1'+yf_2'\frac{\partial z}{\partial x}=0, \qquad ①$$

解得 $$\frac{\partial z}{\partial x}=\frac{f_1'}{yf_2'} \qquad ②$$

①两边再对 x 求导，得

$$f_{11}''-yf_{12}''\frac{\partial z}{\partial x}-yf_{21}''\frac{\partial z}{\partial x}+y^2f_{22}''\left(\frac{\partial z}{\partial x}\right)^2+yf_2'\frac{\partial^2 z}{\partial x^2}=0,$$

$$\frac{\partial^2 z}{\partial x^2}=\frac{1}{yf_2'}\left[-y^2f_{22}''\left(\frac{\partial z}{\partial x}\right)^2+y(f_{12}''+f_{21}'')\frac{\partial z}{\partial x}-f_{11}''\right]$$

祝同学们

考研路上一路顺利！

